

RESCUE JUNIOR - Bill of Materials (BOM)

Team name: CNTROBOT

Hardware							
Number	Bill of Materials	Thai currency	US currency	Note / Primary Source	quantity	Thai currency Total price	US currency Total price
1	Pixy 2.1 CMUcam5	THB 3,200.00	USD 96.97	Official retail price from authorized distributors in Thailand.	1	THB 3,200.00	USD 96.97
2	Lm2596 DC-DC	THB 35.00	USD 1.06	Standard market price for generic electronics components.	1	THB 35.00	USD 1.06
3	Gyro ZX-IMU	THB 390.00	USD 11.82	Official retail price from INEX (Innovative Experiment).	1	THB 390.00	USD 11.82
4	SENSOR SEN002	THB 180.00	USD 5.45	Official retail price from PT-BOT SHOP.	7	THB 1260.00	USD 38.15
5	MG90S servo	THB 75.00	USD 2.27	Average market price for standard metal gear micro servo.	5	THB375.00	USD 11.35
6	Motor PG-300	THB 350.00	USD 10.61	Estimated price per unit from local robotics suppliers.	4	THB 1400.00	USD 42.44
7	POP-32i	THB 2,140.00	USD 64.85	Official retail price for INEX POP-32i Core Board.	1	THB 2,140.00	USD 64.85
8	ZX-Sonar1M	THB 295.00	USD 8.94	Official retail price for INEX Ultrasonic Distance Sensor.	2	THB 590.00	USD 17.88
9	Rubber wheels	THB 700.00	USD 21.21	Official retail price from Prince-BOT SHOP.	4	THB 2800.00	USD 84.88
10	Battery Li-po Helicox 1500mah	THB 475.00	USD 14.40	Official retail price from PT-BOT SHOP.	1	THB 475.00	USD 14.40
11	3D wheels	THB 0.00	USD 0.00	-	4	THB 0.00	USD 0.00
12	Gripper	THB 0.00	USD 0.00	-	1	THB 0.00	USD 0.00
13	Miscellaneous items	THB700.00	USD 21.22	-	1	THB 700.00	USD 21.22
Total price						THB 13,365.00	USD 405.10

Software		
Software / Library Name	Purpose / Function	Author / Owner
pop32i.h (POP32i Library)	Core library for controlling input/output ports, DC motor drivers, and the color TFT LCD display on the POP-32i microcontroller board.	INEX
ZX-IMU Library	A library for reading raw data from the Gyro sensor and configuring orientation and heading calculations for precise robot turning.	INEX
ZX-IMU Library	A custom-built library for interfacing with the Pixy2.1 camera via communication protocols (e.g., I2C, SPI, UART) to read object coordinates, color codes, and line vectors for the rescue robot.	CNTROBOT (Parawat Tumto)
Operating system	Wimdow	